# Answer key Tutors only

In the order of the day: C, D, break, B, A. The numbers the web pages show. "Say" = the one point to make at the debrief. Tutor notes, the plan and the worked SUV example on pages 3-4.

## C Reconstruction and lesion size

1st · simulated · OSEM, 16 subsets

**1–2 SUVmax up, liver mean the same, noise up.**

| No filter | 1 iteration | 3 iterations | 12 iterations |
|---|---|---|---|
| Tumour SUVmax | 11.34 | 12.43 | 13.05 |
| Liver mean | 2.61 | 2.65 | 2.64 |
| Liver noise (SD, % of mean) | 15 % | 23 % | 44 % |

**3 3 iterations: no filter 12.43 · 3 mm 12.01 · 5 mm 10.19 · 7 mm 9.10. Noise 23 → 21 → 13 → 11 %. The filter smooths: less noise, but a lower SUVmax.**

**4 None of them. 9.1 to 13.1 here, 12.49 in the hospital (sharper reconstruction: resolution modelling, time of flight, 2 mm pixels).**

Say: There is no one right SUVmax. Compare SUVs only between scans reconstructed the same way.

**5 Same scanner type and reconstruction? Both EARL-accredited, with the EARL reconstruction? Same uptake time and protocol?**

**1 Small ones lower.**

**2 3 it + 5 mm: 10 mm 36 %, 37 mm 103 %.**

**3 From about 17-22 mm (89-96 %).** Below 2-3 × the blur (6.5 mm) they read low.

**4 Not only one: 5 mm filter with 3 to 12 iterations, and 2 iterations with 3 mm.** No filter, or 7 mm: fails.

**5 About 8** (3 ÷ 0.36), if it is a sphere of even uptake. It could be anywhere above 3.

| Sphere | Recovery | EARL range |
|---|---|---|
| 10 mm | 36 % | 34-57 % |
| 13 mm | 60 % | 59-85 % |
| 17 mm | 89 % | 73-101 % |
| 22 mm | 96 % | 83-109 % |
| 28 mm | 106 % | 91-113 % |
| 37 mm | 103 % | 95-116 % |

Say: Small lesions read low. A tumour that shrinks reads a lower SUVmax even if each cell takes up as much FDG.

## D Contouring on PET

2nd · 3 it, 5 mm filter · expert outline 25.4 mL

**1.1–3 Pairs' own numbers.** A wide window (0-5) gives a bigger outline, a narrow one (0-20) a smaller one.

**2.1 Biggest: SUV 2.5 · smallest: gradient.**

**2.3 SUV 2.5 overlaps the expert best (0.75)**, only because her outline is big and her uptake uneven (a hot core off-centre). It is not a sign that SUV 2.5 is good.

| Method | Threshold | mL | Overlap |
|---|---|---|---|
| 41 % of SUVmax | 4.18 | 7.8 | 0.47 |
| 50 % of SUVmax | 5.09 | 5.1 | 0.33 |
| SUV 2.5 | 2.50 | 15.1 | 0.75 |
| Adaptive | 3.66 | 9.2 | 0.53 |
| Gradient | – | 3.8 | 0.26 |

**2.4 It grows into anything warm next to the tumour. In the liver (background about 2.5) or the brain it fails completely.**

**2.5 No single method. In lung the GTV is contoured on CT, with PET as a guide; an automatic outline is a start that a physician edits.**

### 3 · ANOTHER RECONSTRUCTION

| Reconstruction | SUVmax | 41 % mL | SUV 2.5 mL |
|---|---|---|---|
| 1 iteration, 7 mm | 8.48 | 8.6 | 14.4 |
| 3 iterations, 5 mm | 10.19 | 7.8 | 15.1 |
| 12 iterations, none | 13.05 | 5.5 | 15.4 |

**1 Smaller (7.8 → 5.5 mL).**

**3 The threshold follows the maximum:** a sharper, noisier image → higher SUVmax → 41 % of it is higher → the outline shrinks.

**4 17-37 mm within 10 %.** 13 mm +90 %, 10 mm +394 %: small spheres need higher thresholds.

Say: Change the reconstruction, change the GTV. A PET contour comes with its scanner, its reconstruction and its method.

Simulated scan of FDG-PET-CT-Lesions (autoPET), PETCT_7faf36a152 (Gatidis et al., Sci Data 2022;9:601; TCIA, doi:10.7937/gkr0-xv29; CC BY 4.0). Methods and SUV 2.5: Hatt et al., AAPM TG211, Med Phys 2017;44:e1-e42. EARL1 ranges: Kaalep et al., Eur J Nucl Med Mol Imaging 2018;45:1344-61. Partial volume: Soret et al., J Nucl Med 2007;48:932-45.

## B Attenuation correction

3rd · simulated · 3 iterations, 5 mm filter

### 1 · WITH AND WITHOUT AC

1 **Too high: skin and the body surface, the lungs. Too low: the centre of the body (mediastinum, spine, middle of the liver). (Colour map on: bright = high.)**

2 **Less. Without AC the lung around the tumour reads too high (lung absorbs little), so the tumour stands out less: tumour max ÷ lung around it about 13 with AC, about 8 without.**

Say: Without AC the contrast depends on where the lesion sits, not only on its uptake. Against the deep liver the tumour would even look a little better without AC.

3–4 **Both photons must leave the body. Lines through the middle cross about 30 cm of tissue: only about 1 pair in 18 gets out. Lines through the skin or air-filled lung lose little. No SUV without AC: the values have no unit.**

### 2 · BREATHING

| CT used for AC | Tumour SUVmax | Change |
|---|---|---|
| 2 cm higher (breathed out) | 8.58 | −16 % |
| Matched | 10.19 | – |
| 1 cm lower | 10.19 | 0 % |
| 2 cm lower | 9.42 | −8 % |
| 4 cm lower (deep breath in) | 6.65 | −35 % |

1 **Goes down, to 8.58 (−16 %).** The tumour moves with the lung: the moved CT puts air-filled lung where the PET saw the tumour, so lines through it are corrected too little. Distance from the diaphragm does not protect it.

2 **A curved band at the liver dome, too hot or too cold; the tumour too low. Biggest error (table): 4 cm lower (deep breath in), −35 %.**

3 **On PET on CT: the liver dome and the diaphragm do not line up, a curved cold band under the lung, the PET lesion sitting off the CT mass. Look there every time.**

4 **Shallow breathing for the CT, not a deep breath in · an averaged CT · gated PET with a 4D CT.**

Say: The CT is in the PET. A breathing mismatch moves the SUV with nothing changed in the patient (Erdi 2004: up to 24-30 %).

## A SUV from the DICOM header

4th · real data, the hospital's own reconstruction

### 1 · THE IMAGE

1 **Right lower lobe, posterior, a few cm above the liver dome.**

2 **Bigger. With 0-5, everything above SUV 5 is full black, so the warm edge joins the tumour.**

Say: The window changes what you see, not the data. Remember this when you draw in Block D.

3 **SUVmax 12.49 (the hottest voxel). SUVmean depends on the circle: about 8.6 for a 2 cm circle, 6.1 for 3 cm, 4.4 for 4 cm.**

Say: SUVmax is one voxel; SUVmean is only as good as the circle someone drew.

4 **Heart muscle, kidneys and urine (renal pelvis, bladder), some gut. Normal FDG uptake and excretion, not cancer. (The brain is not shown: the data provider blanked the head.)**

### 2 · THE DICOM HEADER

1 **Weight 70 kg · dose 323 MBq · injection 07:44:00 · half-life 6586.2 s = 109.8 min · Series Time 08:39:01.**

2 **Series Time. Decay Correction = START: the scanner has already corrected every slice back to the series start. Using the slice's Acquisition Time (08:45:35, 6.6 min later) corrects twice: SUVmax 4 % too high.**

3 **Uptake `55.0 min` · fraction left `0.707` · dose at scan start `323 × 0.707 = 228.2 MBq` · SUVmax `40713 Bq/mL ÷ (228200000 Bq ÷ 70000 g) = 40713 ÷ 3260 = 12.49`**

4 **BMI 24.8 · LBM `43.8 kg` · SULmax `12.49 × 43.8 ÷ 70 = 7.8`**

### 3 · WHAT IF?

| Mistake | SUVmax | Change | Why |
|---|---|---|---|
| Weight in pounds (154) | 27.47 | +120 % | SUV is proportional to the weight. |
| Clock 30 min slow (07:14) | 15.09 | +21 % | 85 min of decay assumed, so too little dose left in the calculation. |
| 10 % left in the syringe | 13.88 | +11 % | The screen (12.49) is too low: she got less dose. |

4 **No. The calculation removes the decay exactly, but not the biology: FDG keeps building up in most tumours, so a later scan reads higher. EANM: 60 min (55-75); a repeat scan within 10 min of the first uptake time.**

5 **Weight, dose (and what was left in the syringe), injection and scan times (clocks in sync), uptake time. Also the same scanner and reconstruction (Block C). Blood glucose is not in the DICOM header: ask for it.**

Say: Even with everything matched, SUVmax repeats only to about 12 %. QIBA: a real change from +39 % or −28 %.

SUV: Boellaard et al., EJNMMI 2015;42:328-54. QIBA FDG-PET/CT profile v1.14 (2023). LBM: Janmahasatian et al., Clin Pharmacokinet 2005;44:1051-65. Breathing: Erdi et al., J Nucl Med 2004;45:1287-92; Goerres et al., Radiology 2003;226:906-10; Liu et al., Phys Med Biol 2009;54:7345-62.

# Tutor notes For René

Everything runs in the browser. Nothing to install, no login. Goal: every pair finishes all four blocks; bonus tasks are for fast pairs only.

| Time | Block | Min | Take-home point |
|---|---|---|---|
| 15:00 | Intro: the question; pairs open pet.html (QR code). You show the viewer (3 min) and work one SUV example on screen (5 min): boxes below | 12 | "When you're shown an SUV or a PET contour, what do you need to know before you trust it?" |
| 15:12 | C · Reconstruction and lesion size **(start here)** | 15 | Same patient, different SUVmax |
| 15:27 | D · Contouring on PET | 25 | The method determines the GTV |
| 15:52 | Break | 10 | |
| 16:02 | B · Attenuation correction | 20 | The CT is in the PET |
| 16:22 | A · SUV from the DICOM header | 20 | SUV is a calculation |
| 16:42 | Debrief: the "My answers" page (Block A ends there) and the worksheet; fill in the checklist together | 10 | The checklist |
| 16:52 | 8 min spare | | |

### Intro 1 · Show the viewer (3 min, from the front)

- Open Block C (pet_recon.html). The slider beside the image: **up = towards the head**.
- **Click** a point: its SUV appears under the image. **Drag**: a circle with the SUVmax and SUVmean inside it.
- **Window** 0–5 / 0–10 / 0–20: the SUV at the top of the colour bar. **Colours**: inverted grey (high = dark) or colour (high = bright).
- One task at a time (Next / Previous). Predict boxes lock; then they try it. Each block ends with a button to the next block.

### Intro 2 · One SUV example (5 min, on screen or whiteboard)

Her real DICOM header (Block A). A voxel in the liver stores `8500 Bq/mL`. What is its SUV?

1. Weight 70 kg = `70000 g`. Injected `323 MBq` at `07:44:00`. F-18 half-life `109.8 min` (header: 6586.2 s).
2. Time from injection to scan start (`08:39:01`): `55.0 min`.
3. Fraction left: `2^(−55.0 ÷ 109.8) = 2^(−0.501) = 0.707`.
4. Dose at scan start: `323 MBq × 0.707 = 228.2 MBq = 228200000 Bq`.
5. Per gram of body: `228200000 Bq ÷ 70000 g = 3260 Bq/g`. That is SUV 1 for her.
6. SUV = `8500 Bq/mL ÷ 3260 Bq/g = 2.6` (1 mL of tissue ≈ 1 g, so no unit). A normal liver.

**Say:** In Block A you do this for the tumour's hottest voxel. (The header has two scan times; which one is right is their task A 2.2, so do not name the tag.)

### Before the session

- Open **patrikbrynolfsson.github.io/estro2026-dwi-handson/pet.html** on the room's PC and on your laptop. The QR code on the worksheet goes there.
- The first load takes a few seconds (about 9 MB of images). On the room's Wi-Fi, ask pairs to open it at the start.
- A page that looks broken: Ctrl+F5 once (old copy in the browser).
- Print one worksheet per pair, double-sided. Pairs can also type their answers on the pages.

### How the pages work

- Order C, D, break, B, A: pet.html lists them in that order, with C marked "Start here".
- Answers stay in that browser only; nothing is sent anywhere. "My answers" (top of each page) lists them in the same order, with links back to each task.
- Each block ends with a message box: that is the take-home point.
- Laptops are best; phones work but are cramped.

**Keeping to time**

- Everyone does every block. Pairs behind time: skip the bonus tasks (marked "for fast pairs").
- **Protect D, and its step 3** (another reconstruction): it ties C to the GTV. Do not let C run over 15 min.
- Pairs still in D at the break finish D after it, then go on to B.
- Block A: the time line in step 2 separates uptake time from decay time; "Check my numbers" spots a wrong time.

**Questions you may get**

- **Real or simulated?** Block A is her real scan and DICOM header. C, D and B are a simulated scan built from her real PET and CT, so we can change one thing at a time.
- **Why 10.19 in B–D but 12.49 in A?** A is the hospital's sharper reconstruction; B–D use a standard one. That is Block C's point.
- **Is the expert outline a GTV?** No: the lesion outline from the autoPET dataset, drawn for research, not for radiotherapy.
- **RayStation?** The programme mentions it; these pages replace it, with the same patient case.

## Checklist · suggested answers

| | |
|---|---|
| **1 · SUV calculated right?** | Weight, dose (minus what was left), injection and scan times with synced clocks, uptake time 55–75 min and matched on repeat scans. |
| **2 · Does the CT match the PET?** | Look at PET on CT at the liver dome and around the tumour. Shallow breathing for the CT; no SUV without AC. |
| **3 · Reconstruction and lesion size?** | Same scanner and reconstruction (EARL) on both scans. Lesions under about 2 cm read low. |
| **4 · Which contouring method?** | Which method, threshold and window, on which reconstruction? Check and edit; in lung, contour the GTV on CT. |

Data: FDG-PET-CT-Lesions (autoPET), TCIA, doi:10.7937/gkr0-xv29 (CC BY 4.0). Simulation: the patient's CT → attenuation at 511 keV (Carney et al., Med Phys 2006;33:976–83), blur 6.5 mm, Poisson noise, OSEM with 16 subsets. Numbers read from the web pages on 2 November 2026 (B 1.2 on 10 November 2026).